Multimedia Information Retrieval

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1 Question - 8 pts

Given the documents

- $d_1 = \{\text{it is what it is}\}$
- $d_2 = \{\text{what is it}\}$
- $d_3 = \{\text{it is a pen}\}$

Illustrate the creation of the inverted index, step-by-step. Show the postings list according to the following format:

$$(t_i, n_i) \rightarrow (d_{j_1}, n_{ij_1}), (d_{j_2}, n_{ij_2}) \dots \tag{1}$$

where n_i is the term frequency in the document collection, d_j is a document that contains n_{ij} occurrences of t_i
Let $\mathbf{S}$ denote a 3×3 matrix, where $\mathbf{S}_{j_1,j_2}$ indicates the similarity coefficient between d_{j_1} and d_{j_2}. Compute the entries of $\mathbf{S}$ adopting a suitable tf-idf weighting scheme.

2 Question - 5 pts

Illustrate the DoG detector, explaining why it achieves scale invariance.

3 Question - 5 pts

Illustrate by means of an example the use of a multimedia indexing scheme to compute the result of nearest neighbor queries in a 2-dimensional vector space. Show the execution of the algorithm when either the l_2 or l_1 norm is used to measure distances between vectors.

4 Question - 8 pts

Consider a dataset of N images and a vocabulary of $V = 20$ visual words. The cell array $visualWords$ indicates the visual word associated with each local descriptor of each image. In particular, the cell $visualWords\{i\}$ consists in a vector containing the visual words of the i-th image.

- Load the cell array $visualWords$ from the workspace *bowlsi.mat* and compute the bag-of-words representation of each image. Store the data in a $V \times N$ matrix B

- Consider the 215th image. Show the bag-of-words histogram of the image and find out the visual word vw_{max} having the maximum number of occurrences in the image
- Compute the rank-k approximation to B for $k = 6$ and use LSI to retrieve the images containing the visual word vw_{max}. Rank the images based on cosine similarity and display the ranked list showing for each position the index of the image and the score. Show only results having a cosine similarity score > 0.9

  Example:

  216 0.991

  10 0.983

  130 0.922

  5 0.919

  ...

5 Questions - 6 pts (each answer can be either TRUE or FALSE)

1. Let $A = [\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_n]$, $\mathbf{d}_i \in \mathbb{R}^m$ denote the vector space representation of a document collection. Let $A = USV^T$ denote the SVD of A, and U_K a matrix that includes the first K columns of U.
   - [T] [F] $\mathbf{d}_i = \sum_{k=1}^{K} \sigma_k v_{i,k} \mathbf{u}_k$, $K < n$
   - [T] [F] $U_{i,j}$ indicates the relevance of term i in topic j.
   - [T] [F] $U_{i,j} \geq 0$, $i = 1, \ldots, m$, $j = 1, \ldots, n$.
   - [T] [F] $\mathbf{y}_j = \Sigma_K^{-1} U_K^T \mathbf{d}_j$ is a K-dimensional representation of $\mathbf{d}_j$.
2. Let $\mathbf{M} \in \mathbb{R}^{n \times n}$ denote the PageRank matrix, obtained applying a damping factor m.
   - [T] [F] $\sum_{j=1}^{n} \mathbf{M}_{i,j} = 1$.
   - [T] [F] Power iteration converges to a unique solution if $m > 0$.
   - [T] [F] Dangling nodes can be handled removing the corresponding rows.
   - [T] [F] PageRank score is query dependent.
3. Consider a image search engine that operates on global color descriptors
   - [T] [F] Each image is characterized by a variable length vector.
   - [T] [F] Images are converted to a suitable, perceptually motivated, color model.
   - [T] [F] Increasing the number of bins in a color histogram increases recall and decreases precision.
   - [T] [F] Color descriptors are robust to rotation.